CIE

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TITLE: **Smart Study Planner**

SCHOOL: Chitkara University, Rajpura, Punjab

DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO SHARE LOGIN AND PASSWORD WITH MENTORS

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ABSTRACT

1.1 Problem Background :

It is very hard for students to manage different subjects, assignment deadlines, exams, and their study capacity simultaneously. While most of the traditional planners have been able to incorporate scheduling and managing of tasks, they fail to take into account the factors that impact the capacity to work such as exhaustion, stress levels, sleep, and even motivation levels. The proposed smart planner would be able to incorporate all of these factors to allow for an actual planning and scheduling of studies.

This problem is solved using a coordinated process of making subjects and tasks followed by mental load and lastly creating a schedule. This is facilitated by storing these details on the backend and using them together to make it more realistic. The backend would be able to store these details in MongoDB.

In conclusion, this problem allows for two issues to be addressed at once. These include academic scheduling and student well-being. For instance, if one student has a lot of pending assignments and sleepless nights ahead, this will definitely affect his/her study planning process.

In conclusion, the solution is an attempt at tackling two problems at once, namely academic scheduling and mental health management. If the user finds themselves under pressure due to many unfinished assignments, lack of sleep, difficult subject matter, or deadlines drawing near, this will show through in the system's response while developing the study schedule.

1.2 Objective of the Project :

The objective of the current project is to create a Smart Study Planner that would assist students with subject and task management while creating realistic study schedules depending on several parameters such as deadlines, difficulty level, priority, and mental load. At the moment, the backend of the project has been fully implemented with all major APIs and logic of schedule generation, whereas the frontend has been created in part with all main pages and routing.

One more objective of the project evident from the current implementation is to develop a full-stack system with proper security measures. Specifically, these include OTP-based registration process, password reset functionality, JWT authentication, middleware for validation, central error handler, rate limiting, and documentation with the help of Swagger. Thus, the focus is not only on functionality but also reliability.

As another objective of the project, the translation of subjective student state into planning information needs to be mentioned. In this regard, the system takes stress and motivation levels calculated from parameters such as fatigue, sleep, pending tasks, overdue tasks, workload, and recent completion activity.

1.3 Proposed Solution / System Overview :

The system in question is an online Smart Study Planner with Mental Load Tracking. Users can register, manage subjects, create tasks, store their mental load details, and receive customized schedules. All the parts of the backend (authentications, subjects and tasks management, mental loads recording, analytics endpoint, scheduling algorithms, validation logic, security middleware, API doc using Swagger, automated tests) have been implemented. In the meantime, there are some features implemented on the frontend (React app including such pages as Login, Register, Dashboard, Subjects, Tasks, Mental-load, Schedule, Analytics), even though the frontend implementation may be considered incomplete compared to the backend.

From the architectural point of view, the proposed architecture may be described as full stack and layered. On the frontend side, React with Vite was used along with React router and Axios library. As for the backend part, Node.js and Express were utilized. For persistent storage, MongoDB with Mongoose library was chosen. Finally, the use of protected routes guarantees that each user can only access his/her study data.

The operational mechanism of the system under consideration can be outlined as follows: a student develops subjects, incorporates tasks with due dates and time needed, provides information about fatigue and sleep, and asks for scheduling. The backend calculates priorities, sets a new daily capacity based on the level of fatigue, anticipates fatigue after scheduling studying, and saves the plan for that day, as well as the warnings about overload and study recommendations.

Furthermore, the system provides for data analytics. The previously saved schedules, the state of task completion, and data concerning mental load are used to calculate the number of hours spent studying, fatigue patterns, productivity score, and likelihood of burnout. These results being based on stored academic performance, the system functions not only as a scheduler but as an academic decision-making tool as well.

1.4 Key Features or Functionality :

So far, the main features that have been deployed include user authentication, subject CRUD, task CRUD, mental load logging CRUD, schedule generation from mental load data analysis, analytics APIs, centralized validation and error handling, and API documentation. In terms of the front-end, there are some user interface screens along with the protection of routes for important modules.

On the other hand, when it comes to the authentication module, the project is capable of making a request for OTP for registration, OTP verification, log-in, log-out, fetching the user's profile, updating the user's profile, and resetting the password using OTP from an email. Passwords are hashed using bcrypt and JWT tokens are used for managing user sessions.

In the academic module, a subject is set up by the user, and the tasks are connected to the established subject. Information about the task includes title, deadline, importance, estimated time to complete the task, difficulty, completion status, and schedule details. Tasks can be completed by the user or re-scheduled.

In the planning and analysis module, schedule is created, overloaded days are recognized, fatigue probability is predicted, stress and motivation are estimated, productivity is calculated, and burnout probability is predicted. The best part about this module of the application is that the user gets analytical output from his or her academic activities.

1.5 Technology / Method Used :

For the frontend part, the project makes use of React and Vite, whereas for the backend, it utilizes Node.js, Express, JWT authentication, bcrypt, Nodemailer, Swagger documentation, Jest, Supertest testing, and scheduling. For database management, MongoDB and Mongoose are used. With all the above technologies in play, the existing prototype can be described as one that emphasizes the backend more compared to the frontend part, which is not yet completely developed.

With regards to the methodology of the system, it is important to emphasize that it is not based on complex machine learning algorithms but rather on heuristics with weights. It implies that various parameters such as the proximity of deadlines, importance of subjects, complexity of assignments, their expected completion time, fatigue levels, hours of sleep, and past load are taken into account and used in certain formulas and rules.

One of the key equations employed is the Task Priority Score which is used in scheduling tasks. Using a simplified formula, Priority Score equals Deadline Urgency plus 1.4 multiplied by Subject Priority plus 1.2 multiplied by Task Difficulty plus 0.8 multiplied by Estimated Duration plus 1.6 multiplied by Task Priority. Deadline Urgency is obtained by 10 divided by number of days left before deadline. Therefore, it ensures that important and urgent tasks are automatically scheduled at the top of the queue.

Fatigue-based capacity control is another key approach used in the algorithm. When fatigue is extremely high, the daily study capacity is reduced to 0.55 times the actual capacity, while when it is high, it will be 0.75, but when the fatigue level is low, it increases to 1.15. Once the tasks are scheduled, fatigue level can be predicted based on the formula: Fatigue Prediction equals Fatigue Level plus Allocated Hours divided by 2 plus (Stress minus Motivation) divided by 4.

Mathematical approximation methods are applied to compute mental states and burnout probability. Stress and motivation come from fatigue, sleep, unfinished tasks, overdue tasks, difficult tasks, accomplished tasks, and overloaded days. The computation of burnout probability takes into account weighted normalized values of average fatigue, average stress, lack of motivation, sleep penalty, unfinished workload, overloaded days, accomplished workload penalty, and planned working hours.

1.6 Expected Outcome & Benefits :

The expected result of the implementation of the proposed solution would be the creation of a tool for effective management of studying. Due to the current results of the project, the backend component of the application is ready and ready to support authentication, scheduling, analysis, and mental load evaluation functions. The frontend of the project is also ready, consisting of the basic page and navigation elements, however, it should not be assumed that this part of the project is finished as well, as not all features have been implemented.

The obvious advantage of such an application for students would be its ability to help organize one’s studies. Without the need to choose which subject to begin studying first, how many hours should be spent on studying and whether it was a particularly heavy day, the person would receive a plan already taking into account all the factors mentioned.

Another advantage is increased understanding of one’s own academic wellbeing. Since the platform contains data on fatigue, sleep, stress prediction, motivation prediction, and burnout symptoms, it will provide more insight into how performance affects one’s psychological state. Thus, this project can be used for purposes other than just planning; it may also serve to improve one’s studying habits through self-regulation.

From the academic perspective, the current project represents full-stack development, RESTful API creation, database design, scheduling algorithms, weighted analysis, and user management. As a result, the intended outcome includes not only the practical application of the product but also its significance as a software engineering project.

1.7 Original Contribution Statement :

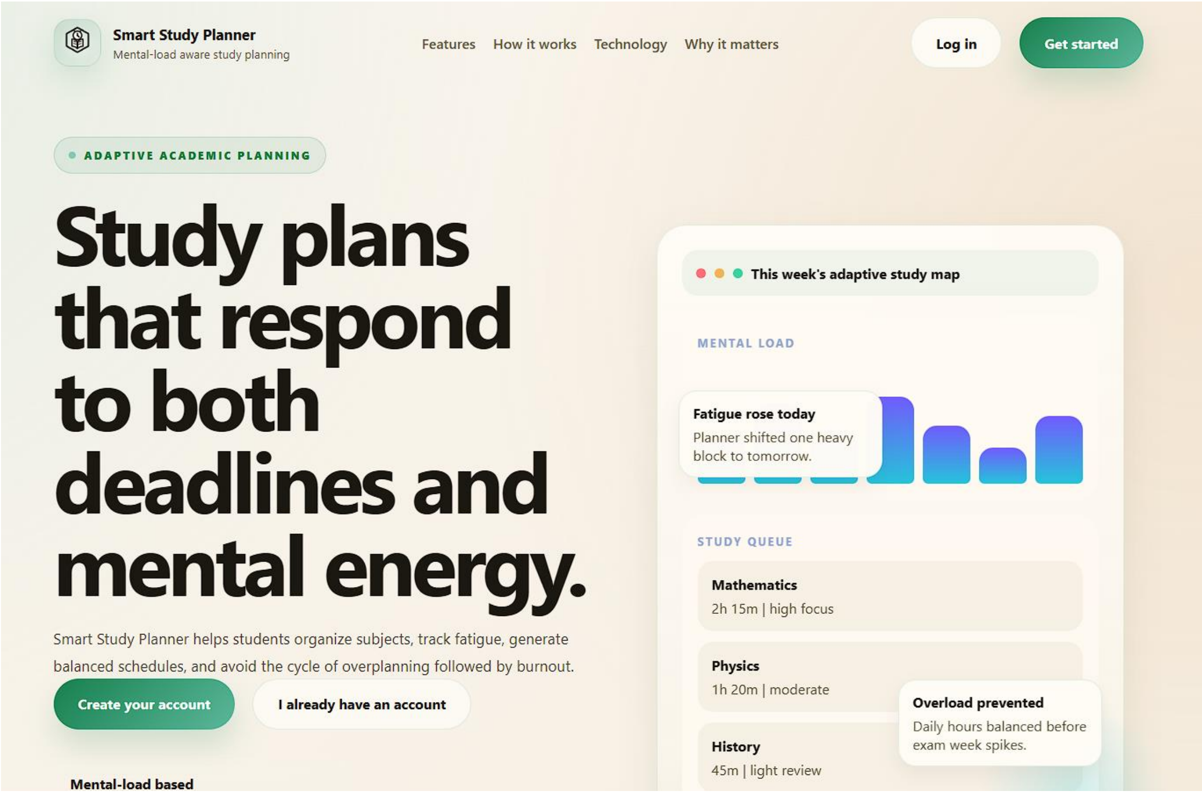
The originality of this project is in the integration of mental fatigue into the process of scheduling rather than in using deadlines alone to schedule tasks. At present, the project adds the creation of a back-end architecture for the smart academic planner featuring adaptive scheduling, overloading detection, analytical features, and secure APIs, in addition to a prototype of the front-end through which students will work with the application.

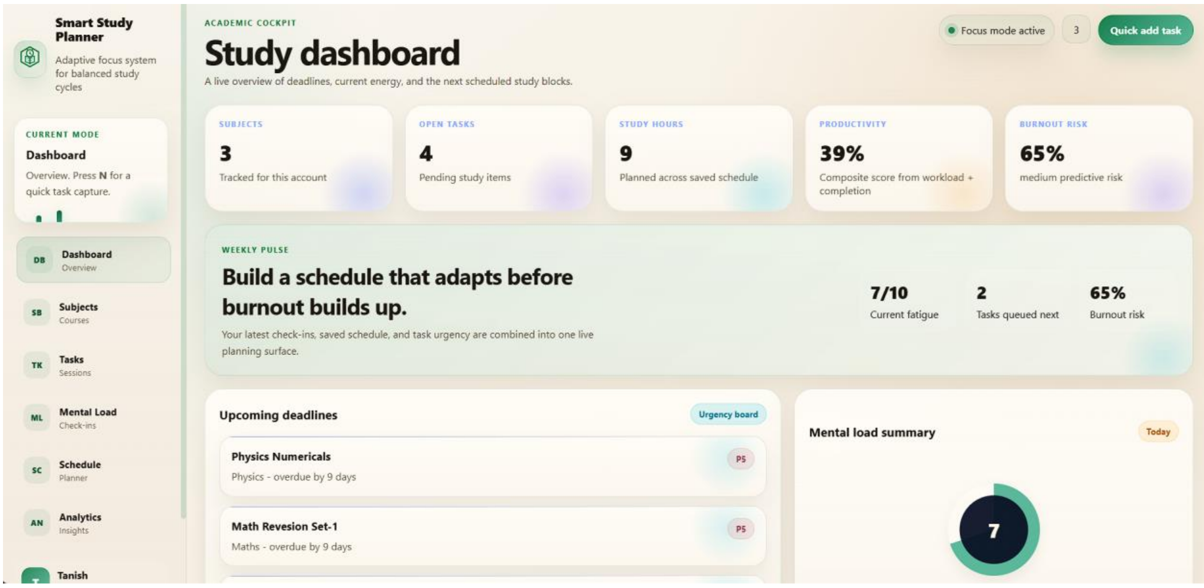
Originality in this project is observed in the incorporation of various student signals within one engine for planning. The main feature distinguishing this planner from other planners is in the connection between the urgency of tasks, the importance of subjects, the fatigue level, sleep quality, stress, motivation, and burnout detection.

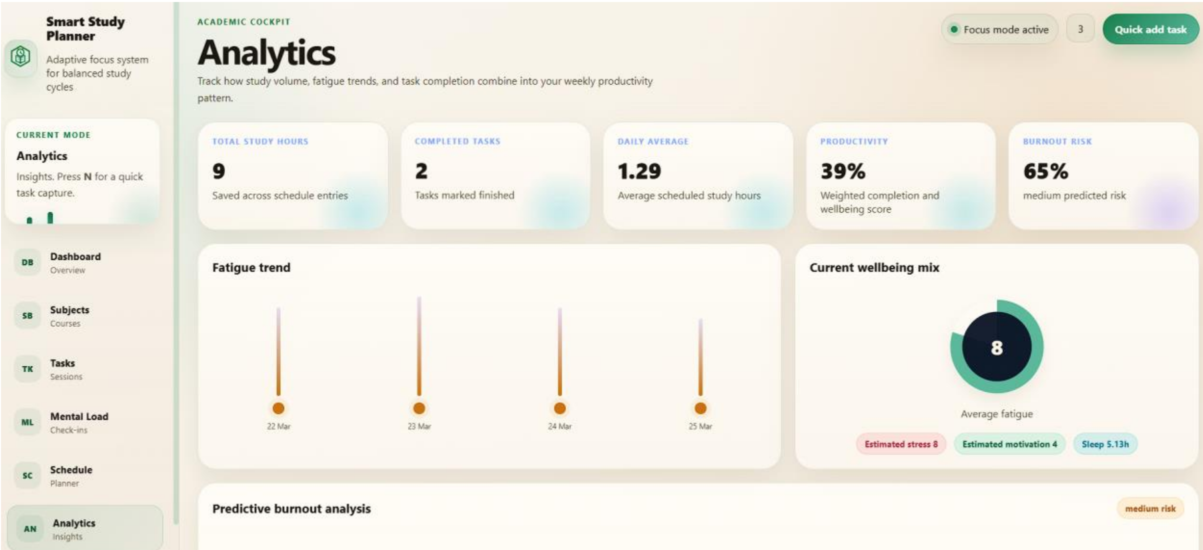
An additional contribution of this nature is that of the interpretive heuristic framework adopted by the backend. This project is no "black box" whose workings are inscrutable to reviewer, mentor, and end-user alike; it makes use of weighted equations and logic-based thresholds, thus allowing its operations to be interpreted. This is incredibly useful for a project within an academic context, as all the reasoning involved can be accounted for.

The project is also unique in terms of its development approach, having incorporated elements such as authentication, validation, testing, and Swagger documentation into a modular system architecture along with a use case involving study planning. Thus, this project is not only novel conceptually but technologically as well.

PROJECT CODE SNIPPETS / SCREENSHOTS






Smart Study Planner
Adaptive focus system for balanced study cycles
CURRENT MODE
Analytics
Insights. Press N for a quick task capture.
DB Dashboard Overview
SB Subjects Courses
TK Tasks Sessions
ML Mental Load Check-ins
SC Schedule Planner
AN Analytics Insights
ACADEMIC COCKPIT
Analytics
Track how study volume, fatigue trends, and task completion combine into your weekly productivity pattern.
Focus mode active
3
Quick add task
TOTAL STUDY HOURS
9
Saved across schedule entries
COMPLETED TASKS
2
Tasks marked finished
DAILY AVERAGE
1.29
Average scheduled study hours
PRODUCTIVITY
39%
Weighted completion and wellbeing score
BURNOUT RISK
65%
medium predicted risk
Fatigue trend
22 Mar
23 Mar
24 Mar
25 Mar
Current wellbeing mix
8
Average fatigue
Estimated stress 8
Estimated motivation 4
Sleep 5.13h
Predictive burnout analysis
medium risk